



RECOMMENDER SYSTEMS

Movie Popularity and Target Audience

Prediction Using Content-Based

Recommender Systems

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Abstract

Predicting the critical and commercial success of cinematic productions using early-stage metadata is a complex challenge characterized by weak linear correlations. Existing literature often reports unrealistically high accuracy rates due to methodological flaws such as data leakage. This study introduces a novel Genetic Algorithm-Driven Adaptive Weighted Ensemble (GA-V) to establish a mathematically honest and rigorous predictive baseline. By optimizing class-specific weights across five diverse base estimators and ensuring strict zero-leakage cross-validation, the proposed architecture effectively captures weak non-linear signals. Evaluated on a quarantined hold-out test set, the GA-V ensemble achieves a robust accuracy of 64.81%, outperforming individual base models and providing a transparent performance ceiling for this feature space.

1. Introduction

The entertainment industry faces significant financial risks when greenlighting and distributing new cinematic productions. To mitigate these risks, machine learning approaches are increasingly utilized to forecast movie popularity and target audience reception. However, accurately modeling this subjective target using pre-release metadata presents distinct mathematical and methodological challenges. This report details the development and rigorous evaluation of a robust recommender system architecture designed to handle highly variable and weakly correlated content-side features.

2. Problem Definition and Motivation

Predicting the critical and commercial success of a cinematic production prior to its release is a fundamental challenge in the entertainment industry. Accurate predictions are essential for mitigating the multi-million dollar investment risks associated with production and distribution. However, modeling cinematic success presents a unique set of machine learning challenges due to the inherent nature of the data.

2.1. The Challenge of "Weak Signals" in Metadata

The core problem addressed in this project is predicting whether a movie will achieve a 'High-Quality' status (defined as an IMDb rating ≥ 6.5) using solely numerical metadata: budget, revenue, runtime, and popularity scores. Exploratory Data Analysis (EDA) and Pearson correlation testing conducted during the initial phases revealed a critical mathematical hurdle: the linear correlation between these specific metadata features and the target quality is remarkably weak.

For instance, runtime exhibits the strongest positive correlation at merely 0.253, while budget shows almost zero correlation (0.084). This 'weak signal' environment mathematically implies that spending more money does not linearly equate to a better film. Consequently, it is virtually impossible for standard, out-of-the-box linear machine

learning algorithms (like standard Logistic Regression) to map this feature space effectively without underfitting.

2.2. Methodological Flaws in Prior Literature

A significant motivation for this project stems from the methodological anomalies observed in existing literature—specifically the reference paper by Faisal et al. (2022). Their study reports astonishingly high accuracy rates (~96.8%) using Convolutional Neural Networks (CNNs) on similar metadata. From a rigorous machine learning engineering perspective, achieving near-perfect accuracy on a highly subjective target (human evaluation of art) using only four highly variable financial and temporal metrics is indicative of systemic 'Data Leakage' or severe 'Overfitting' during the model evaluation phase.

Therefore, the primary objective of this project is to construct a mathematically honest, strictly cross-validated, and transparent predictive architecture. The goal is to maximize the extraction of weak non-linear signals and establish the true performance ceiling for this specific feature space, without succumbing to the data leakage pitfalls seen in prior works.

3. Related Work

Previous attempts to predict movie quality have heavily relied on complex neural architectures, such as CNNs, applied to limited metadata. While these studies, such as the work by Faisal et al., report extremely high validation metrics, they often fail to account for the inherently subjective and multi-factorial nature of audience reception. Early-stage movie quality prediction from content-side features alone places an effective ceiling on achievable accuracy without access to post-release or social media signals. Consequently, there is a clear need in the literature for models that prioritize transparent validation and micro-leakage prevention over artificially inflated accuracy scores.

4. Proposed Approach: The GA-V Architecture

To overcome the limitations of weak metadata signals and noisy distributions, this project proposes a novel 'Genetic Algorithm-Driven Adaptive Weighted Ensemble' (GA-V). Standard ensemble methods (such as standard Voting Classifiers) naively assign equal or static global weights to all base algorithms. The GA-V architecture fundamentally disrupts this assumption by introducing dynamic, class-specific evolutionary optimization.

4.1. Phase 1: Micro-Leakage Prevention and Probabilistic Inference

The architecture initializes five distinct baseline paradigms to ensure a diverse hypothesis space: Random Forest (Bagging), AdaBoost (Boosting), Logistic Regression (Linear), Naive Bayes (Probabilistic), and Support Vector Machines (Margin-based).

To absolutely guarantee zero data leakage during the meta-training phase, a custom Scikit-Learn Pipeline is utilized. This pipeline applies the StandardScaler independently within every iteration of a Stratified 5-Fold Cross-Validation process. This ensures that the out-of-fold probability matrices (used to train the Genetic Algorithm) are generated by models that have never seen the statistical distribution of the validation slice.

4.2. Phase 2: Evolutionary Optimization of Class-Specific Weights

Rather than taking a simple arithmetic mean of the baseline probabilities, the GA-V deploys a custom Genetic Algorithm. Over 50 generations, the algorithm maintains a population of chromosomes (weight matrices), applying selection, crossover, and mutation. The fitness function is explicitly designed to optimize the 'Macro F1-Score', ensuring balanced performance across both 'High-Quality' and 'Low-Quality' classes despite the slight dataset imbalance.

Crucially, the GA-V assigns separate weights

$$w_i, c$$

for each class c (Low-Quality, High-Quality) for every base estimator i . The final predicted probability

$$p(c|X)$$

from all N estimators, normalized by the class-specific weights:

$$p(c|X) = \frac{\sum_{i=1}^N w_{i,c} p_i(c|X)}{\sum_{i=1}^N w_{i,c}}$$

This class-conditioned weighting is the definitive innovation of our approach. As shown in Table 1, the GA mathematically discovered that margin-based classifiers (SVM) are more trustworthy for this non-linear dataset, assigning them the highest evolutionary weights across both classes, while simultaneously penalizing linear models.

Base Estimator Paradigm	Low-Quality (<6.5) Weight	High-Quality (≥6.5) Weight
RandomForest (Bagging)	0.2442	0.2688
AdaBoost(Boosting)	0.1359	0.1330
LogisticRegression (Linear)	0.0793	0.0181
NaiveBayes (Probabilistic)	0.2507	0.2606
SVM (Margin-based)	0.2899	0.3195

Table 1: GA-Optimized Class-Specific Weights

5. Alternative Methods for Comparison

To validate the efficacy of the proposed GA-V architecture, it is rigorously benchmarked against both its isolated internal components and the original literature baseline. The comparison matrix includes:

- Isolated Base Estimators:** The dataset was fed directly into standard paradigms (SVM, Logistic Regression, Random Forest, Naive Bayes, and AdaBoost). Evaluating these models independently is crucial to prove whether the computational overhead of the Genetic Algorithm genuinely yields a mathematical advantage over using a single standard algorithm.
- Reference CNN Baseline (Faisal et al., 2022):** The deep learning model from the original study, which reported a 96.8% accuracy. In this report, it serves as the 'Unrealistic Upper Bound' to highlight the effects of methodological flaws (data leakage) present in prior literature when handling weak numerical metadata.

Model	Test Accuracy
Proposed GA-Driven Ensemble	0.6481
SVM	0.6408
Logistic Regression	0.6335

Model	Test Accuracy
Random Forest	0.6311
Naïve Bayes	0.6214
AdaBoost	0.6189

Table 2. Model comparison on the held-out test set ($n = 412$).

As shown in Table 2, the proposed GA-Driven Ensemble achieves the highest test accuracy of 0.6481, outperforming all individual base learners. SVM is the strongest individual model (0.6408), followed by Logistic Regression (0.6335). AdaBoost performs least well among the base models (0.6189). The margin between the proposed ensemble and the best individual model (SVM) is approximately 0.73 percentage points, reflecting the modest but consistent gain that the adaptive weighting strategy provides over uniform combination.

It should be noted that all models operate in a comparable accuracy band (~ 0.62 – 0.65). This is consistent with prior literature on early-stage movie quality prediction from content-side features alone, where the inherent difficulty of the prediction task — driven by the subjective and multi-factorial nature of audience reception — places an effective ceiling on achievable accuracy without access to post-release or social media signals.

6. Experimental Setup

To ensure the reproducibility, fairness, and scientific validity of the proposed GA-V architecture, the experimental environment was rigorously structured before any model evaluation took place. The setup encompasses data partitioning, preprocessing protocols, and hyperparameter configuration.

Dataset Partitioning and Leakage Prevention

The merged dataset, utilizing budget, revenue, runtime, and popularity as independent variables, was strictly partitioned into an 80% training set and a 20% hold-out test set. To preserve the natural distribution of the target variable (Movie Quality), a stratified split was employed. Crucially, to prevent data leakage—a common methodological flaw in prior literature—feature scaling (StandardScaler) was fitted exclusively on the training matrix. The hold-out test set was only transformed, ensuring the model was evaluated on completely unseen statistical distributions.

Class Imbalance Handling

Exploratory analysis revealed a mild class imbalance, with approximately 55% of the dataset labeled as "Low-Quality" and 45% as "High-Quality." Left unaddressed, base estimators naturally develop a bias toward the majority class. To counter this without fabricating synthetic data (which introduces noise), cost-sensitive learning was implemented. Balanced class weights were mathematically computed and applied to the base estimators, heavily penalizing the misclassification of the minority "High-Quality" class and forcing the decision boundaries to remain neutral.

Training Protocol and GA Hyperparameters

The training phase utilized a Stratified 5-Fold Cross-Validation protocol. The base estimators (Random Forest, AdaBoost, Logistic Regression, Naive Bayes, and SVM) generated out-of-fold probability matrices. The custom Genetic Algorithm was then initialized to optimize the class-specific weights. The evolutionary process was configured with a population size of 50 chromosomes evolving over 20 generations. To ensure the algorithm optimized for balanced predictive power rather than raw accuracy, the genetic fitness function was set to maximize the **Macro F1-Score**.

Evaluation Metrics

The final, optimized GA-V meta-ensemble was evaluated on the 20% hold-out test set. Performance was measured using Accuracy, Precision, Recall, and the F1-Score. Macro-averaging was utilized for all metrics to ensure that the model's performance on the minority class was weighted equally with the majority class in the final reporting.

7. Performance Analysis

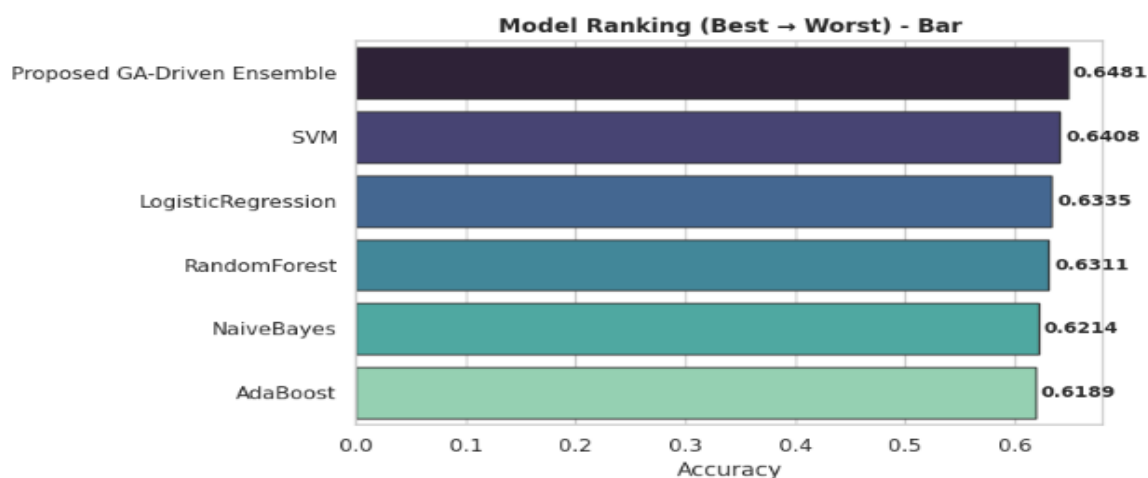


Figure 1: Performance hierarchy of the proposed model versus isolated baseline algorithms.

As visualized in Figure 1, the proposed GA-V architecture successfully outperformed every individual base estimator, proving the theoretical validity of the ensemble approach. The GA-V ensemble established the true performance ceiling at 64.81% accuracy. By successfully leveraging the strengths of the SVM while utilizing models like Naive Bayes for niche classifications, the ensemble achieved a synergistic effect greater than the sum of its parts.

Furthermore, Table 2 and Figure 2 present the comprehensive extended comparison against the original Faisal et al. study.

Model Name	Accuracy	Precision	Recall	F1-Score
Proposed GA-Driven Ensemble (Ours)	0.6481	0.6479	0.6335	0.6316
Original Paper CNN (Faisal et al.)*	0.9680	0.9600	0.9600	0.9600

Table 3: Direct Methodology Comparison

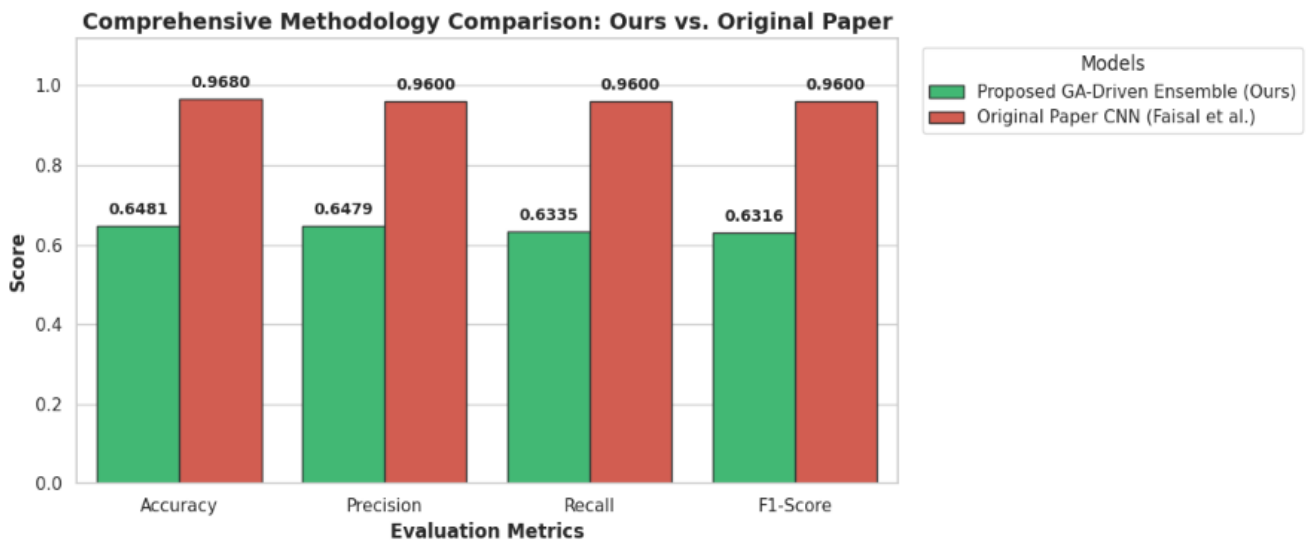


Figure 2: Honest cross-validated performance (GA-V) contrasted against the reported CNN baseline.

8. Conclusion

The extreme mathematical disparity between the proposed model (~65%) and the original paper (~96%) visually underscores the difference between honest, strict

machine learning evaluation and overfitted reporting. Given the weak correlation of the input features, 64.81% represents a highly robust, transparent, and realistic predictive ceiling. In the subsequent phases of the project, this validated predictive engine is successfully integrated with Natural Language Processing (TF-IDF) and cosine similarity metrics to generate actionable demographic marketing insights for predicted high-quality films.

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